Aadittya Tiwari

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Objective

Experienced undergrad with skills ranging from Quantum Computing programming, Machine Learning, and Finite Element Analysis looking for a quantum computing algorithms research role for Summer 2026.

Education

Georgia Institute of Technology | Atlanta, GA *January 2026 – Present*
Bachelor of Science in Electrical Engineering, GPA: TBD Expected Graduation, August 2027

University of California, Santa Cruz | Santa Cruz, California *June 2024 – December 2025*
Transfer with 108/180 Quarter system Credit Hours, GPA 3.95

Skills

Programming Languages and Frameworks: Python, Pytorch, Pennylane, Qiskit, Pandas, MATLAB, C, C++
Platforms: COMSOL, SLURM, VS Code, Git, Supabase, Microsoft Office

Experience

University of California, Los Angeles | TensorWorks Lab *November 2024 – December 2025*
Third Author Researcher

- Tasked with applying my experience with Quantum Machine Learning (QML) models to help a PhD Student and a Visiting Professor with analyzing advantages/disadvantages of using QML in industrial turbine degradation forecasting Guided the research on 3 new SiGe devices that resulted in $400K of additional funding and 2 patents.
- Architected a novel Machine Learning model that uses a Quantum Long Short Term memory (QLSTM) as the Generator and classical Convolutional Neural Network as discriminator for a Generative Adversarial Network (GAN) for time-series forecasting tasks
- Implemented Quantum GAN to forecast vibration signal's acceleration values within a degrading turbine, converging in 2.7x less epochs (49 classical epochs, 18 quantum) with only 169 QLSTM parameters VS 8,186,113 classical LSTM parameters. Code open-sourced on github with 14 stars and 4 forks
- Deployed repository to a High Performance Computing CPU cluster using SLURM to parameterize train/test split across 5 values and repeat each split 10 times to quantify variance in each model's RMSE performance per split, per model
- Paper accepted and presented to ESREL'24 civil engineering conference Summer 2024. Currently revising conference paper for journal submission with guidance of Professor and PhD student.

University of California, Santa Cruz | Applied Photonics Group *May – August 2016*
First Author researcher

- Tasked with calculating electric field distribution within a solid-state nanopore in order to determine optimal structure to maximize nanopore chip's capture radius of particles
- Recreated exact nanopore chip structure used by PhD student in lab within COMSOL for finite element analysis of the electric field and voltage drop distribution
- Eliminated out of memory errors by creating custom mesh with varying element sizes depending on proximity to regions of interest and editing COMSOL's internal files to increase access to RAM
- Automated parameter sweep to vary the amount of nanopores (1,2,4,8,10) and distance between nanopores (100 nanometers to 6 micrometers in 100 nm increments) to calculate capture radius for each configuration
- Exported raw electric field distribution data from COMSOL to MATLAB to find an equation to express particle capture radius as a logistic function of nanopore count and placement

Relevant Coursework

Embedded Systems with C, Optoelectronics & Photonics, Circuit Analysis, Digital Design and Assembly Language, Modern Physics